

Richard James Telford

Department of Biological Sciences | University of Bergen

📍 Postboks 7803, 5020 Bergen, Norway

✉ richard.telford@uib.no

🆔 0000-0001-9826-3076

☎ +47 94124571

🌐 richardjtelford.github.io

🌐 richard-telford

🐦 @richardjtelford

🔗 richardjtelford

🔗 Google Scholar

Employment History

Associate Professor of Plant Ecology

University of Bergen

Bergen, Norway

2007 -- present

Postdoctoral Researcher (Forsker II)

Bjerknes Centre for Climate Research

Bergen, Norway

2003 -- 2006

Research Associate and Teaching Fellow

Newcastle University

Newcastle, U.K.

2000 -- 2002

Research Associate and Teaching Fellow

Lancaster University

Lancaster, U.K.

1998 -- 2000

Education

Certificate in Learning and Teaching in Higher Education

Newcastle University

Newcastle, U.K.

2000 -- 2001

Ph.D. in Earth Sciences

University of Wales, Aberystwyth

Aberystwyth, Wales

1995 -- 1998

- "Diatom stratigraphies of Lakes Awassa and Tilo, Ethiopia: Holocene records of groundwater variability and climate change." Supervised by Dr H. F. Lamb.

B.A. (Hons) 2.1 Natural Sciences (Plant Sciences)

Clare College, Cambridge University

Cambridge, U.K.

1991 -- 1994

- Secondary subjects: Geology and Ecology

Research

Research Interests

Palaeoecology; Quantitative palaeoecological reconstruction; Climate change; Climate impacts; Reproducible science

Publications

ISI Citations: 7880

H-Index: 43

Number Papers: 108

Number Chapters: 8

Number Packages: 3

Journal Articles

Althuizen, I. H. J., Gya, R., Jaroszynska, F., Lee, H., **Telford, R. J.**, Chipperfield, J., Enquist, B. J., Goldberg, D. E., and Vandvik, V. (2026) Climate-induced shifts in plant investment strategies regulate ecosystem carbon cycling across alpine grasslands. *Journal of Ecology* 114, e70364. doi 10.1111/1365-2745.70364

Gaudard, J., **Telford, R. J.**, Chacon-Labela, J., Dawson, H. R., Enquist, B. J., Töpper, J. P., and others. (2025) fluxible: An R package to process ecosystem gas fluxes from closed-loop chambers in an automated and reproducible way. *Methods in Ecology and Evolution* 16, 2560-2568. doi 10.1111/2041-210x.70161

Gould, E., Fraser, H. S., Parker, T. H., Nakagawa, S., and others including **Telford, R. J.** (2025) Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology. *BMC Biology* 23, 35. doi 10.1186/s12915-024-02101-x

Vandvik, V., Halbritter, A. H., Macias-Fauria, M., Maitner, B. S., Michaletz, S. T., **Telford, R. J.**, and others. (2025) Plant traits and associated ecological data from global change experiments and climate gradients in Norway. *Scientific Data* 12, 1477. doi 10.1038/s41597-025-05509-4

Halbritter, A. H., Vandvik, V., Cotner, S. H., Farfan-Rios, W., and others including **Telford, R. J.** (2024) Plant trait and vegetation data along a 1314 m elevation gradient with fire history in Puna grasslands, Perú. *Scientific Data* 11, 225. doi 10.1038/s41597-024-02980-3

Jaroszynska, F., Lie Olsen, S., Gya, R., Klanderud, K., **Telford, R. J.**, and Vandvik, V. (2024) Plant functional group interactions intensify with warming in alpine grasslands. *Ecography* 2024, e07018. doi 10.1111/ecog.07018

Jaroszynska, F., Althuizen, I., Halbritter, A. H., Klanderud, K., Lee, H., **Telford, R. J.**, and Vandvik, V. (2023) Bryophytes dominate plant regulation of soil microclimate in alpine grasslands. *Oikos* 2023, e10091. doi 10.1111/oik.10091

Lynn, J. S., Gya, R., Klanderud, K., **Telford, R. J.**, Goldberg, D. E., and Vandvik, V. (2023) Traits help explain species' performance away from their climate niche centre. *Diversity and Distributions* 29, 962-978. doi 10.1111/ddi.13718

Maitner, B. S., Halbritter, A. H., **Telford, R. J.**, Strydom, T., Chacon, J., Lamanna, C., and others. (2023) Bootstrapping outperforms community-weighted approaches for estimating the shapes of phenotypic distributions. *Methods in Ecology and Evolution* 14, 2592-2610. doi 10.1111/2041-210x.14160

Vandvik, V., Halbritter, A. H., Althuizen, I. H. J., Christiansen, C. T., and others including **Telford, R. J.** (2023) Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard. *Scientific Data* 10, 578. doi 10.1038/s41597-023-02467-7

Velle, L. G., Haugum, S. V., **Telford, R. J.**, Thorvaldsen, P., and Vandvik, V. (2023) Prescribed burning can promote recovery of Atlantic coastal heathlands suffering dieback after extreme drought events. *Applied Vegetation Science* 26, e12760. doi 10.1111/avsc.12760

Cao, X., Chen, J., Tian, F., Xu, Q., Herzs Schuh, U., **Telford, R. J.**, Huang, X., Zheng, Z., Shen, C., and Li, W. (2022) Long-distance modern analogues bias results of pollen-based precipitation reconstructions. *Science Bulletin* 67, 1115-1117. doi 10.1016/j.scib.2022.01.003

Herzs Schuh, U., Böhmer, T., Li, C., Cao, X., Hébert, R., Dallmeyer, A., **Telford, R. J.**, and Kruse, S. (2022) Reversals in Temperature-Precipitation Correlations in the Northern Hemisphere Extratropics During the Holocene. *Geophysical Research Letters* 49, e2022GL099730. doi 10.1029/2022gl099730

Strømme, C. B., Lane, A. K., Halbritter, A. H., Law, E., and others including **Telford, R. J.** (2022) Close to open-Factors that hinder and promote open science in ecology research and education. *PLoS ONE* 17, e0278339. doi 10.1371/journal.pone.0278339

Vandvik, V., Althuizen, I. H. J., Jaroszynska, F., Krüger, L. C., and others including **Telford, R. J.** (2022) The role of plant functional groups mediating climate impacts on carbon and biodiversity of alpine grasslands. *Scientific Data* 9, 451. doi 10.1038/s41597-022-01559-0

Geange, S. R., Oppen, J. von, Strydom, T., Boakye, M., and others including **Telford, R. J.** (2021) Next-generation field courses: Integrating Open Science and online learning. *Ecology and Evolution* 11, 3577-3587. doi 10.1002/ece3.7009

Lynn, J. S., Klanderud, K., **Telford, R. J.**, Goldberg, D. E., and Vandvik, V. (2021) Macroecological context predicts species' responses to climate warming. *Global Change Biology* 27, 2088-2101. doi 10.1111/gcb.15532

Moros, M., Deckker, P. D., Perner, K., Ninnemann, U. S., Wacker, L., **Telford, R. J.**, Jansen, E., Blanz, T., and Schneider, R. (2021) Hydrographic shifts south of Australia over the last deglaciation and possible interhemispheric linkages. *Quaternary Research* 102, 130-141. doi 10.1017/qua.2021.12

Thomson, E. R., Spiegel, M. P., Althuizen, I. H. J., Bass, P., and others including **Telford, R. J.** (2021) Multiscale mapping of plant functional groups and plant traits in the High Arctic using field spectroscopy, UAV imagery and Sentinel-2A data. *Environmental Research Letters* 16, 055006. doi 10.1088/1748-9326/abf464

Carlson, C. J., Chipperfield, J. D., Benito, B. M., **Telford, R. J.**, and O'Hara, R. B. (2020) Don't gamble the COVID-19 response on ecological hypotheses. *Nature Ecology & Evolution* 4, 1155-1155. doi 10.1038/s41559-020-1279-2

Carlson, C. J., Chipperfield, J. D., Benito, B. M., **Telford, R. J.**, and O'Hara, R. B. (2020) Species distribution models are inappropriate for COVID-19. *Nature Ecology & Evolution* 4, 770-771. doi 10.1038/s41559-020-1212-8

Chevalier, M., Davis, B. A., Heiri, O., Seppä, H., and others including **Telford, R. J.** (2020) Pollen-based climate reconstruction techniques for late Quaternary studies. *Earth-Science Reviews* 210, 103384. doi 10.1016/j.earscirev.2020.103384

Contina, A., Yanco, S. W., Pierce, A. K., DePrenger-Levin, M., and others including **Telford, R. J.** (2020) Comment on Å global-scale ecological niche model to predict SARS-CoV-2 coronavirus infection rate. *Ecological Modelling* 436, 109288. doi 10.1016/j.ecolmodel.2020.109288

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- Bouchet, V. M., **Telford, R. J.**, Rygg, B., Oug, E., and Alve, E. (2018) Can benthic foraminifera serve as proxies for changes in benthic macrofaunal community structure? Implications for the definition of reference conditions. *Marine Environmental Research* 137, 24-36. doi 10.1016/j.marenvres.2018.02.023
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- Perner, K., Moros, M., Deckker, P. D., Blanz, T., Wacker, L., **Telford, R. J.**, Siegel, H., Schneider, R., and Jansen, E. (2018) Heat export from the tropics drives mid to late Holocene palaeoceanographic changes offshore southern Australia. *Quaternary Science Reviews* 180, 96-110. doi 10.1016/j.quascirev.2017.11.033
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Book Chapters

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R packages

Telford, R. J. (2026) *checker: Checks R Configuration Set Up Correctly Before Class*. R package version 0.1.5. CRAN 0.1.5

Telford, R. J., Halbritter, A. H., and Maitner, B. S. (2023) *traitstrap: Bootstrap Trait Values to Calculate Moments*. R package version 0.1.0. CRAN 0.1.0

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Funding

PalaeoDrivers: Quantifying the Drivers of Palaeoecological Change (1400k NOK)

PI

Research Council of Norway

2012-2016

Teaching

Teaching Interests

Open and Reproducible Science; Quantitative Palaeoecology; Multivariate Statistics; Biostatistics with R

Courses Taught

Bio300b Biostatistics University of Bergen	<i>Course leader</i> 2022 -- present
Bio302 Methods for Open and Reproducible Science University of Bergen	<i>Course leader</i> 2025 -- present
Bio303 Ordination and Gradient Analysis University of Bergen	<i>Course leader</i> 2007 -- present
Bio250 Palaeoecology University of Bergen	<i>Team member</i> 2007 -- present
Bio940 Open, Reproducible and Transparent Science in Ecology University of Bergen/Living Norway	<i>Team member</i> 2022 -- present
Plant Functional Trait Course University of Bergen/University of Arizona	<i>Team member</i> 2015 -- 2023
Bio302 Biological data analysis II University of Bergen	<i>Course leader</i> 2007 -- 2024
R for Palaeoecologists University of Cape Town	<i>Course leader</i> 2019
Recent Environmental Change University of Newcastle	<i>Temp course leader</i> 2002
Holocene Environmental Change University of Lancaster	<i>Temp course leader</i> 2000
Physical Geography of the British Isles University of Lancaster	<i>Temp course leader</i> 2000

Workshops Taught

Reproducible data analysis Københavns Universitet	<i>Workshop leader</i> 2018
Age-depth modelling 2nd COST INTIMATE training school	<i>Session leader</i> 2014
Age-depth modelling ECORD summer school	<i>Session leader</i> 2013

Supervision

Postdoc

Amy Eycott Matrix Project	<i>Co-supervisor</i> 2009 -- 2015
Mathias Trachsel PalaeoDrivers Project	<i>Supervisor</i> 2012 -- 2016

Ph.D

Nadine Michaela Arzt

Mechanisms underlying the success of range-expanding species under climate change and their impacts on biodiversity and ecosystem functioning

Co-supervisor

2023 -- Ongoing

Cecilie Iden Nilsen

Acoustic telemetry of salmonids in lakes

Co-supervisor

2025 -- Ongoing

Mari Lie Larsen

Evidence-based policymaking: Exploring the effectiveness of salmon aquaculture management in Norway

Co-supervisor

2022 -- 2025

Siri Vatsø Haugum

Land-use and climate impacts on drought resistance and resilience in coastal heath-land ecosystems

Co-supervisor

2016 -- 2021

Shad Kenneth Mahlum

From the fjords to the rivers: Evaluating the spatial distribution of escaped farmed salmon to inform ecologically relevant management strategies

Co-supervisor

2015 -- 2020

Perpetra Akite

Spatial and matrix influences on the biogeography of insect taxa in forest fragments in central Uganda

Co-supervisor

2009 -- 2015

Collins Bulafu

Diversity and reproductive traits of woody species in forest fragments in and around Kampala area, Uganda

Co-supervisor

2009 -- 2014

Masters

Kirsti Rindal

Brain-gut-microbiota-axis study

Co-supervisor

2023 -- 2024

Moss Tesfayesus

Mangrove forest extent and status along the Eritrean coast

Co-supervisor

2013 -- 2014

Sofie Söderström

Interspecific whale associations in the Norwegian Seas

Supervisor

2011 -- 2012

Gunnar Kvifte

MATRIX: Psychodidae of Budongo Forest

Supervisor

2009 -- 2011

Moreen Uwimbabazi

MATRIX: Bird diversity in forest fragments around Budongo Forest

Co-supervisor

2008 -- 2010

Kristoffer Hauge

MATRIX: Bat biodiversity in tropical forest fragments

Supervisor

2008 -- 2010

Therese Kronstad

MATRIX: Butterfly biodiversity in Ugandan tropical forest fragments

Supervisor

2008 -- 2010

Wang Manfei

Population dynamics of trees colonizing coastal heathlands on Lurøy

Co-supervisor

2007 -- 2008

Professional Memberships & Service

Reviewing

African Journal of Ecology, Aquaculture, Boreas, Dendrochrologia, Ecography, Ecology and Evolution, Ecology Letters, Environmental Science and Pollution Research International, Flora, Geophysical Research Letters, Global Change Biology, Global Ecology and Biogeography, Journal of Biogeography, Journal of Paleolimnology, Methods in Ecology and Evolution, Nature Communications, Nature Geoscience, PloS one, Science of the Total Environment, Scientific Reports, The Holocene, The International Journal of Sustainable Development and World Ecology

Skills & Software

Methods

Quantitative palaeoecology, age-depth modelling, trait-based ecology, biostatistics

Software

R (expert), shiny (expert), Git (proficient), Quarto (expert), R package development (expert)